

ICAD RT-4 · Set 4 — Answer Sheet

Exam 12-10-2026 · 75 Q · 300 marks · 3 hours · +4 correct, -1 wrong, 0 blank

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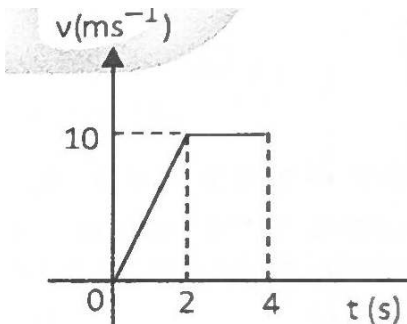
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Physics

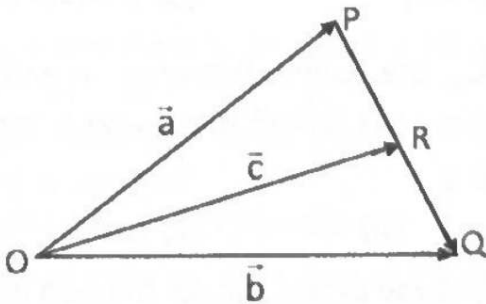
PART A • Q1–20 • SINGLE CORRECT

1. P2L6V5Q3 Car speeds up uniformly $0 \rightarrow 10$ m/s in 2 s then stays constant; distance in the first two seconds. (a) 20 m (b) 5 m (c) 10 m (d) 2 m

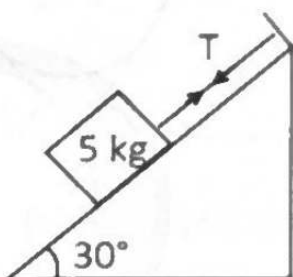


2. P3L1V5Q10 A point $P(x,y)$ has position vector $\mathbf{r} = x\hat{i} + y\hat{j}$. Which of the following are correct? (One or more correct) (a) If $P(3,4)$, then $|\mathbf{r}| = 5$ (b) If $P(0,0)$, then $|\mathbf{r}| = 0$ (c) If $P(2,2)$, then $|\mathbf{r}| = 2$ (d) For any $P(x,y)$, $|\mathbf{r}| = \sqrt{x^2 + y^2}$

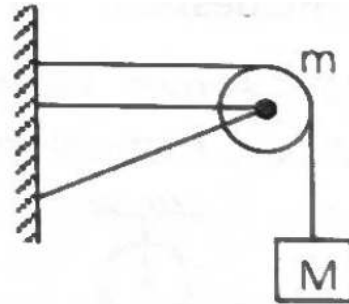
3. P3L2V3Q5 Figure shows three vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, where R is the mid-point of PQ . Then which of the following relations is correct? (a) $\mathbf{a} + \mathbf{b} = 2\mathbf{c}$ (b) $\mathbf{a} + \mathbf{b} = \mathbf{c}$ (c) $\mathbf{a} - \mathbf{b} = 2\mathbf{c}$ (d) $\mathbf{a} - \mathbf{b} = \mathbf{c}$



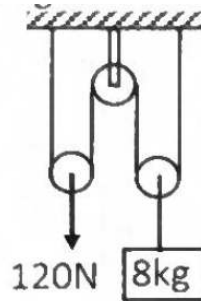
4. P4L3V5Q10 A block at rest slides down a smooth inclined plane which makes an angle 60° with the vertical and reaches the ground in t_1 seconds. Another block is dropped vertically from the same point and reaches the ground in t_2 seconds. The ratio of $t_1:t_2$ is: (a) 1:2 (b) 2:1 (c) 1:3 (d) $1:\sqrt{2}$
5. P4L4V6Q6 A body is thrown upwards with velocity 100 m/s and it travels 5 m in the last second of its upward journey. If the same body is thrown upward with velocity 200 m/s, what distance will it travel in the last second of its upward journey? (a) 5 (b) 10 (c) 20 (d) 25 m
6. P4L5V3Q1 If $\mathbf{a} = 4\mathbf{t}$, find $\mathbf{v}$ and then the displacement after 2 seconds. (a) 9 (b) 5.33 m (c) 10.67 m (d) 16 m
7. P4L12V3Q2 A boat steers perpendicular across a 180 m wide river with $v_{br} = 6$ m/s, $v_r = 4$ m/s. How far downstream does it land? (a) 60 m (b) 80 m (c) 120 m (d) 100 m
8. P5L2V4Q1 A 5 kg block rests on a 30° incline, held by a string with tension T along the incline; tension in the string is: (a) 5 N (b) 25 N (c) 10 N (d) 0



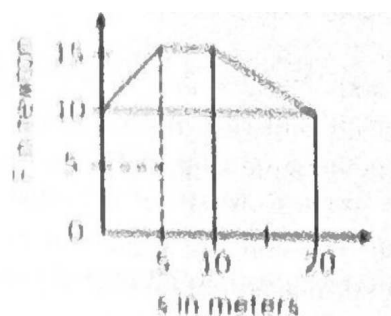
9. P6L3V4Q4 A string of negligible mass over a clamped pulley of mass m supports a block of mass M . Force on the pulley by the clamp: (a) $\sqrt{2} Mg$ (b) $\sqrt{2} mg$ (c) $\sqrt{(M+m)^2 + m^2} g$ (d) $g\sqrt{(M+m)^2 + M^2}$



10. P6L4V5Q4 If the string is pulled down with a force 120 N as shown (movable-pulley system), then acceleration of the 8 kg block would be. (a) 10 (b) 5 (c) 0 (d) 4 m/s²

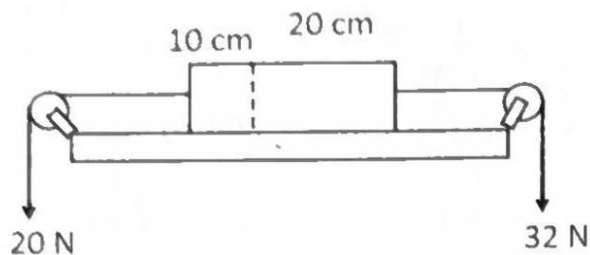


11. P6L4VSuggQ2 A smooth wedge has angle $\theta = 30^\circ$. Find acceleration of a block sliding down it. (a) 5.0 (b) 8.7 (c) 3.0 (d) 9.8 m/s²
12. P7L1V3Q5 A 1 kg block rests on a rough horizontal surface ($\mu_s = 0.5$, $\mu_k = 0.3$). It is pulled to the right by a time-varying force $F(t) = 1 + t$ N for $0 \leq t \leq 6$ s ($g = 10$ m/s²). Which is correct? (a) The block remains at rest until $t = 3$ s. (b) The block starts sliding when $F(t) = 5$ N. (c) While sliding, the frictional force is 5 N. (d) During sliding, applied force and friction act in the same direction.
13. P8L3V4Q7 A box of mass 25 kg starts from rest and slides down an inclined plane 8 metre long and 5 metre high. It is found to move at the bottom at 7 m/sec. What is the force of friction? (a) 79.6 N (b) 96.6 N (c) 76.6 N (d) 116.6 N
14. P8L3V5Q3 Figure shows the frictional force versus displacement for a particle in motion. The loss of kinetic energy in travelling over $s = 0$ to 20 m will be: (graph: f rises from 10 N at $s = 0$ to 15 N at $s = 5$ m, stays 15 N up to $s = 10$ m, then falls linearly to 10 N at $s = 20$ m) (a) 250 J (b) 200 J (c) 150 J (d) 10 J



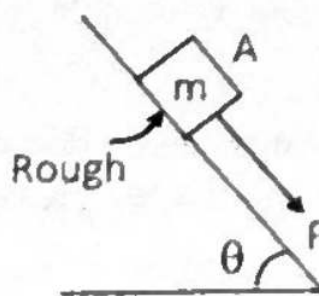
15. P8L8V5Q7 A car weighing 1000 kg is going up an incline with a slope of 2 in 25 at a steady speed of 18 kph. If $g = 10 \text{ ms}^{-2}$, the power of its engine is (a) 4 kW (b) 50 kW (c) 625 kW (d) 25 kW

16. P6L4V5Q3 Figure shows a uniform rod of length 30 cm, mass 3.0 kg. The strings are pulled by constant forces 20 N and 32 N. Find the force exerted by the 20 cm part of the rod on the 10 cm part (in Newton). All surfaces smooth; strings and pulleys light. (write the value)



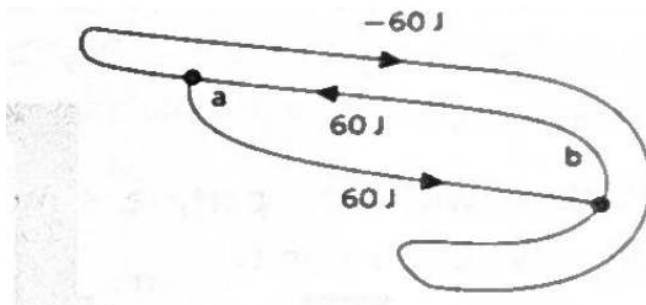
17. P8L8V4Q1 A hoist lifts a mass of 100 kg with a constant acceleration of $a = 1 \text{ m/s}^2$. Find the power (in watts) of the hoist after $t = 2 \text{ s}$. ($g = 10 \text{ m/s}^2$) (write the value)

18. P7L1V4Q2 Find the number of forces in the F.B.D. of block 'A': block A of mass m on a rough inclined plane of angle θ , with an external force F applied to it. (write the value)



19. P8L2V2Q3 A body is constrained to move in the y -direction. It is subjected to a force $(-2\hat{i} + 15\hat{j} + 6\hat{k})$ newton. What is the work done (in J) by this force in moving the body through a distance of 10 m? (write the value)

20. P8L5V3Q1 The figure shows three paths connecting points 'a' and 'b'. A single force F does the indicated work on a particle moving along each path in the indicated direction — the three paths carry -60 J , 60 J and 60 J . On the basis of this information, is force F conservative? If your answer is YES mark (1); if your answer is NO mark (2). (write the value)

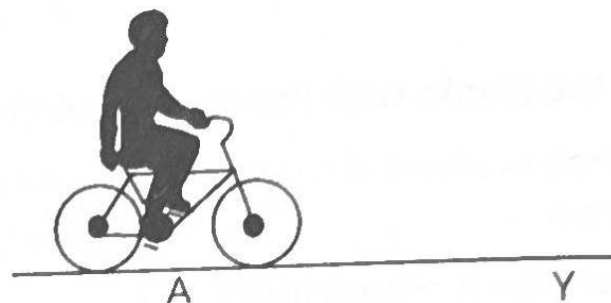


PART B · Q21–25 · NUMERICAL VALUE

21. P2L4V4Q10 Paragraph: $f(x) = (x-1)(x+2)^2 - x$ at local max/min & their values

22. P3L5V5Q2 A spy plane is tracked by radar. At $t=0$ its position is (100m, 200m, 1000m); 130s later, (2500m, 1200m, 1000m). Find a unit vector along the plane's velocity and the magnitude of its average velocity. (unit vector (12/13, 5/13, 0); 20 m/s)

23. P4L2V3Q1 A cyclist accelerates uniformly from rest at A and reaches Y at 9 ms^{-1} , covering the last 10 m of the ride in 2 s. Determine the acceleration (m/s^2)



24. P2L2V5Q4 If $y = (3x+4)/(4x+5)$, find dy/dx at $x = -1$

25. P4L1V6Q4 A body starts at $x=0$ with instantaneous velocity $v(t) = v_0(1 - t/\tau)$ for $0 \leq t \leq \tau$, then rests. Compute the total distance travelled (two decimal places, in metres)

Chemistry

PART A · Q1–20 · SINGLE CORRECT

- C1L7V4Q10 Four gases (He, N₂, O₂, CO₂) same T — correct order of u_{mp} . (a) $\text{CO}_2 > \text{O}_2 > \text{N}_2 > \text{He}$ (b) $\text{He} > \text{N}_2 > \text{O}_2 > \text{CO}_2$ (c) $\text{N}_2 > \text{He} > \text{O}_2 > \text{CO}_2$ (d) $\text{He} > \text{O}_2 > \text{N}_2 > \text{CO}_2$
- C2L5V3Q4 Determine excitation energy from $n=1$ to $n=3$ for hydrogen. (a) 12.1 eV (b) 10.2 eV (c) 9.4 eV (d) 8.5 eV
- C2L5V3Q5 Convert total energy -3.4 eV to joules. (a) $-5.44 \times 10^{-19} \text{ J}$ (b) $-3.4 \times 10^{-19} \text{ J}$ (c) $-2.18 \times 10^{-19} \text{ J}$ (d) $-1.7 \times 10^{-19} \text{ J}$

- C2L9V6Q10 Choose correct statements (shape $\leftrightarrow$ l, orientation $\leftrightarrow$ m, energy $\leftrightarrow$ n, degeneracy $\leftrightarrow$ l). (a) The shape of an atomic orbital depends upon azimuthal quantum number (b) The orientation of an atomic orbital depends upon the magnetic quantum number (c) The energy of an electron in an atomic orbital of multi-electron atom depends upon principal quantum number only (d) The number of degenerate atomic orbitals of one type depends upon the value of azimuthal quantum number

5. C3L2V4Q5 4 g-atom of Ag contains: (108 g / 4 g / 432 g / None)

6. C3L3V4Q3 A gaseous mixture contains oxygen and nitrogen in the ratio 1 : 4 by weight. The ratio of their number of molecules is (a) 1:4 (b) 1:8 (c) 7:32 (d) 3:16
7. C3L4V3Q3 Empirical formula from N = 87.5%, H = 12.5%. (a) NH₂ (b) NH₃ (c) NH (d) N₂H₃
8. C3L4V5Q5 0.1 mole carbohydrate (EF = CH₂O) contains 1 g H. Molecular formula? (a) C₅H₁₀O₅ (b) C₆H₁₂O₆ (c) C₄H₈O₄ (d) C₃H₆O₃
9. C4L3V6Q4 Assertion (A): For the isoelectronic ions O²⁻, F⁻, Na⁺ and Mg²⁺, the radius of Mg²⁺ is the smallest. Reason (R): Ionic size decreases in an isoelectronic series as nuclear charge increases. (a) Both A and R are true, and R is the correct explanation of A (b) Both A and R are true, but R is NOT the correct explanation of A (c) A is true but R is false (d) A is false but R is true
10. C4L4V4Q2 IE₁, IE₂ and IE₃ values are 100, 150 and 1500 eV respectively. The element can be: (a) Na (b) B (c) Be (d) F
11. C4L5V6Q4 Arrange the following electronic configurations in order of increasing negative electron gain enthalpy: (i) 1s²2s²2p⁶3s²3p⁵ (ii) 1s²2s²2p⁵ (iii) 1s²2s²2p³ (iv) 1s²2s²2p⁶3s¹ — (a) (iii) < (iv) < (ii) < (i) (b) (ii) < (iii) < (iv) < (i) (c) (iv) < (iii) < (i) < (ii) (d) (iii) < (ii) < (iv) < (i)
12. C5L1V4Q5 Lattice energy of an ionic compound depends upon: (a) charge on the ions only (b) size of the ions only (c) packing of the ions only (d) charge and size of the ions

PART B · Q21–25 · NUMERICAL VALUE

21. C1L2V4Q5 0.200 L at 1 atm, 0 °C — volume at 236.14 °C, same P. (a) 373 mL (b) 400 mL (c) 460 mL (d) 546 mL
22. C2L8V5Q2 Electron at v=2.05×10⁷ m/s — wavelength (in m)
23. C5L2V5Q10 How many of the following molecules do not follow the Lewis octet rule: BeCl₂, BF₃, PCl₅, SF₆, SCl₂, IF₇, CCl₄?
24. C1L1V6Q3 A/R: barometer moved coastal→hill, Hg column falls. (a) Both (A) and (R) are true and (R) is the correct explanation of (A). (b) Both (A) and (R) are true but (R) is not the correct explanation of (A). (c) (A) is true but (R) is false. (d) (A) is false but (R) is true.
25. C1L4V3Q2 2 mol A + 3 mol B, total 500 mm — partial pressure of A. (a) 200 mm (b) 300 mm (c) 400 mm (d) 250 mm

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M1L1V4Q7 Sum of two angles = 1 radian, difference = 1°; smallest angle. (a) $(90/\pi - 1/2)^\circ$ (b) $(90/\pi + 1/2)^\circ$ (c) $(180/\pi - 1)^\circ$ (d) $(180/\pi + 1)^\circ$
2. M2L2V4Q2 $(A-B) \cup (B-A)$ equals (a) $(A \cap B) \cup (A \cup B)$ (b) $(A \cup B) - (A \cap B)$ (c) $A - (A \cap B)$ (d) $(A \cup B) - B$
3. M3L1V2Q2 A, B, C, D of a quadrilateral $\rightarrow \tan((A+B)/4) - \cot((C+D)/4)$. (a) 0 (b) 1 (c) -1 (d) 2
4. M4L3V4Q9 For which y does $|x - [x]| = y$ have a solution? (a) [0,1] (b) (0,1] (c) [0,1] (d) (0,1)
5. M4L4V5Q3 Domain of $f(x) = \sec x$. (a) $R \setminus \{(2n+1)\pi/2, n \in Z\}$ (b) $R \setminus \{n\pi, n \in Z\}$ (c) $R \setminus \{n\pi/3, n \in Z\}$ (d) R
6. M5CL1V6Q5 Given both θ and ϕ are acute angles and $\sin\theta = 1/2$, $\cos\phi = 1/3$, then the value of $\theta + \phi$ belongs to: (a) $(\pi/3, \pi/2)$ (b) $(\pi/2, 2\pi/3)$ (c) $(2\pi/3, 5\pi/6)$ (d) $(5\pi/6, \pi)$
7. M5BL2V4Q5 If $\tan(\alpha) = x/(x+1)$ and $\tan(\beta) = 1/(2x+1)$, then $\alpha + \beta$ is equal to (a) $\pi/2$ (b) $\pi/3$ (c) $\pi/6$ (d) $\pi/4$
8. M5CL3V5Q9 $1/(\tan 3A - \tan A) - 1/(\cot 3A - \cot A) =$ (a) $\tan A$ (b) $\tan 2A$ (c) $\cot A$ (d) $\cot 2A$
9. M6L2V5Q5 If $(16 - x^2)\sqrt{3 - x} = 0$, then 'x' is: (a) $\{-4, 3\}$ (b) $\{-3, 4\}$ (c) $\{\pm 4, 3\}$ (d) none of these
10. M6L5V6Q7 If $((2 - x^2)(x - 3)^3)/((x + 1)(x^2 - 3x - 4)) \geq 0$, then x belongs to: (a) $[-\sqrt{2}, 0) \cup (-1, \sqrt{2}] \cup [3, 4)$ (b) $[-\sqrt{2}, 0) \cup (-1, \sqrt{2}]$ (c) $[-\sqrt{2}, 0) \cup (-1, \sqrt{2}]$ (d) $[-\sqrt{2}, -1) \cup (-1, \sqrt{2}] \cup [3, 4)$
11. M7L2V4Q7 $1/(\log_{\sqrt{bc}} abc) + 1/(\log_{\sqrt{ca}} abc) + 1/(\log_{\sqrt{ab}} abc)$ has the value equal to (a) $1/2$ (b) 1 (c) 2 (d) 4
12. M7L4V3Q4 If $\log_{10} 5 = x$, then $\log_5 1250$ equals: (a) $3 - 1/x$ (b) $2 + 1/x$ (c) $3 + 1/x$ (d) $2 - 1/x$
13. M8BL1V4Q9 If a, b, c are in A.P. as well as in G.P. then (a) $a = b \neq c$ (b) $a \neq b = c$ (c) $a \neq b \neq c$ (d) $a = b = c$
14. M8BL1V5Q4 The common difference of an A.P. whose first term is unity and whose second, tenth and thirty fourth terms are in GP is (a) $1/5$ (b) $1/3$ (c) $1/6$ (d) $1/9$
15. M8CL1V3Q2 The sum of three numbers in H.P. is 37 and the sum of their reciprocals is $1/4$. Find the numbers. (a) 10, 12, 15 (b) 10, 12, 16 (c) 12, 14, 16 (d) None of these
16. M5AL1V4Q9 If $A + B = 135^\circ$, then $(1 + \cot A)(1 + \cot B) = ?$ (a) 1 (b) 2 (c) 3 (d) 4
17. M8AL2V3Q3 The number of terms of the A.P. 3, 7, 11, 15 ... to be taken so that the sum is 406 is (a) 5 (b) 10 (c) 12 (d) 14
18. M8BL3V3Q1 The third term of a G.P. is 4. Find the product of First Five terms. (write the value)
19. M8AL3V3Q1 If $a_1 = a_2 = 2$, $a_n = a_{n-1} - 1$ ($n > 2$), then a_5 is (write the value)

- 20.** M8BL3V3Q3 The numbers of bacteria in a certain culture doubles every hour. If there were 15 bacteria present in the culture originally, how many bacteria will be present at the end of 4th hour. (write the value)

PART B · Q21–25 · NUMERICAL VALUE

- 21.** M2L2V5Q10 A, B have 99 common elements $\rightarrow$ common elements of $A \times B$ & $B \times A = 121\lambda^2 \rightarrow |\lambda|$
- 22.** M7L4V6Q3 If the solution set of $(2/5)^{((6-5x)/(2+5x))} < 25/4$ is $(-\infty, a) \cup (a/5, \infty)$, then the value of 2^{-a} is:
- 23.** M5AL2V3Q1 $\cos^2 75^\circ - \cos^2 15^\circ = x$, then $\sqrt{3} x = ?$
- 24.** M6L1V4Q10 If $1/(x-1) \leq 2$, then find the non-negative integer which does not satisfy the inequality.
- 25.** M8AL1V4Q1 If 10 times the 10th term of an A.P. is equal to 15 times the 15th term, then the 25th term of this A.P. is